

Yao He

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🌐 [shockwaveHe](https://shockwavehe.github.io) | 🌐 [yaohe09](https://yaohe09.github.io) | 🌐 shockwavehe.github.io | 🌐 Yao He

Education

Stanford University

M.S. in Electrical Engineering

- GPA: 4.060/4.0

Stanford, CA

2024-2026 (expected)

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)

B.E. in Electronic Information

- First Class Honor
- MGPA: 3.864/4.0 (2/44), CGPA: 3.783/4.0 (6/144)
- Scholarship & Award: 2020-2021 CUHK-Shenzhen Excellent Student Award (Top 1%), Academic Performance Scholarship (2019-2021 for Top 5%), Dean's Lists of School of Science and Engineering (2019-2022), Shaw College Master's List (2019-2021 for Top 5%), Bowen Scholarship (2018-2022)

Shenzhen, Guangdong, China

2018-2022

Publications

Chronologically Ordered, *=Equal Authorship

- [1] **Repurposing 2D Diffusion Models for 3D Shape Completions**
Yao He, YoungJoong Kwon, Tiange Xiang, Wenxiao Cai, Ehsan Adeli
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (Under Review)
- [2] **Locomotion Beyond Feet**
Tae Hoon Yang*, Jiacheng Hu*, Haochen Shi*, Zhicong Zhang, Daniel Jiang, Weizhuo Wang, Yao He, Zhen Wu, Yifan Hou, Monroe Kennedy, Shuran Song, Karen Liu
IEEE International Conference on Robotics and Automation (ICRA), (Under Review)
- [3] **Robust Visual Odometry using Rigidly-Bundled Arbitrarily-Arranged Multi-Cameras**
Huai Yu, Junhao Wang, Yao He, Wen Yang, Gui-Song Xia
IEEE Robotics and Automation Letters, 2025
- [4] **Robot Trains Robot: Automatic Real-World Policy Adaptation and Learning for Humanoids**
Kaizhe Hu*, Haochen Shi*, Yao He, Weizhuo Wang, C. Karen Liu, Shuran Song
Conference on Robot Learning (CoRL), 2025
- [5] **Feed-forward Human Performance Capture via Progressive Canonical Space Updates**
YoungJoong Kwon, Yao He, Hee Jung Choi, Chen Geng, Zhengmao Liu, Jiajun Wu, Ehsan Adeli
The Fourteenth International Conference on Learning Representations (ICLR), 2026
- [6] **Artist-Created Mesh Generation from Raw Observation**
Yao He, Youngjoong Kwon, Wenxiao Cai, Ehsan Adeli
International Conference on Computer Vision (ICCV), 2025, End-to-End 3D Learning Workshop
- [7] **Stem-OB: Generalizable Visual Imitation Learning with Stem-Like Convergent Observation through Diffusion Inversion**
Kaizhe Hu*, Zihang Rui*, Yao He, Yuyao Liu, Pu Hua, Huazhe Xu
The Thirteenth International Conference on Learning Representations (ICLR), 2025, Spotlight.
- [8] **Demonstrating ViSafe: Vision-Enabled Safety for High-Speed Detection and Avoid**
Parv Kapoor*, Ian Higgins*, Nikhil Keetha*, Jay Patrikar*, Brady Moan, Zelin Ye, Yao He, Ivan Cisneros, Yaoyu Hu, Changliu Liu, Eunsuk Kang, Sebastian Scherer
Robotics Science and Systems (RSS), 2025
- [9] **FiReStereo: Forest InfraRed Stereo Dataset for UAS Depth Perception in Visually Degraded Environments**
Devansh Dhrafani*, Yifei Liu*, Andrew Jong, Ukcheol Shin, Yao He, Tyler Harp, Yaoyu Hu, Jean Oh, Sebastian Scherer
IEEE Robotics and Automation Letters, 2025
- [10] **FoundLoc: Vision-based Onboard Aerial Localization in the Wild**
Yao He*, Ivan Cisneros*, Nikhil Keetha, Jay Patrikar, Zelin Ye, Ian Higgins, Yaoyu Hu, Parv Kapoor, Sebastian Scherer
IEEE Robotics and Automation Letters, (Under Review)
- [11] **Towards Robust Visual-Inertial Odometry with Multiple Non-Overlapping Monocular Cameras**
Yao He, Huai Yu, Wen Yang, Sebastian Scherer
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022
- [12] **Computational Efficient Simulation of Kelvin Wake for Unmanned Surface Vehicles**
Yao He, Qinbo Sun, Weimin Qi, Xiaoqiang Ji, Huihuan Qian
IEEE International Conference on Real-time Computing and Robotics (RCAR), 2021

Experience

The Movement Lab (TML), Robotics and Embodied Artificial Intelligence Lab (REAL)

Member of *Toddlerbot team* advised by Prof. C. Karen Liu and Prof. Shuran Song

Stanford, CA

Jan 2025 – Present

Stanford Translational AI Lab (STAI), Stanford Vision & Learning Lab (SVL)

Graduate research student advised by Prof. Ehsan Adeli and Prof. Youngjoong Kwon

Stanford, CA

Sep 2024 – Present

Tsinghua Embodied AI Lab, Institute for Interdisciplinary Information Sciences, Tsinghua University

Mentored by Prof. Huazhe Xu

Beijing, China

June 2024 – Sept 2024

AirLab, Carnegie Mellon University Research Assistant Advised by Prof. Sebastian Scherer <i>Robotics Institute Summer Scholar (RISS)</i> advised by Prof. Sebastian Scherer and Prof. Huai Yu	Pittsburgh, PA Nov. 2022 – May 2024 May 2021 – Aug. 2021
Robotics & Artificial Intelligence Laboratory, CUHK-Shenzhen Undergraduate Research Intern Mentored by Prof. Huihuan Qian	Shenzhen, Guangdong, China Sept. 2020 – May 2021

Leadership & Service

Reviewer IJRR, RA-L, IROS, ICRA, CVPR	2021 - Present
CMU MRSD Team Mentor <i>Mentoring CMU MRSD students to work on hands-on robotics topics proposed by an industry partner or CMU lab</i> Team: EmberEye Resilient Subcanopy UAS Navigation Through Smoke for Wildfire Applications	Sept. 2023 - May. 2024
Tartan Planning Series <i>Organizer</i> Organized an interactive series of talks, tutorials, and learning on planning for robotics with world-renowned pioneers	Mar. 2023 - May. 2023
ICCV SLAM Challenge <i>Organizer</i> - Provide datasets TartanAir and SubT-MRS, aiming to push the robustness of SLAM algorithms in challenging environments - My work on Multispectral Inertial Odometry (MSO) is presented at SLAM Challenge Summary	Mar. 2023 - May. 2023
Teaching <i>Teaching Assistant</i> Robot Perception, EE/CS 227, Fall 2025	Stanford University
<i>Undergraduate Student Teaching Fellow</i> - Introduction to Computer Science: Programming Paradigms, CSC3002, Spring 2021 - Ordinary Differential Equations, MAT 2002, Spring 2022 - Calculus II, MAT1002, Spring 2020 - Mechanics, PHY1001, Fall 2019	CUHK-Shenzhen

Skills

Programming Languages: Python, C/C++, CUDA, Matlab, Julia
Libraries: OpenCV, Eigen, Ceres, NVIDIA VPI, Pytorch, PCL, Numpy
Environments & Tools: Linux, ROS, Docker, CMake, Conda, UE4/AirSim, IsaacSim, Git, Jupyter